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SUNIL GUPTA
CO-FOUNDER & CEO,
YOTTA DATA SERVICES



“The average size of the data centres increased from 20–30 MW range to 50–100 MW... Those that are able to support high power requirement of AI, are being built now”

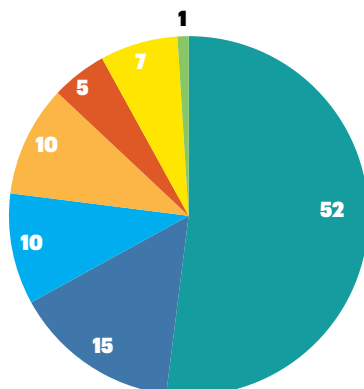
VIVEK DAHIYA
MD & HEAD-DATA CENTRE
ADVISORY TEAM,
ASIA PACIFIC,
CUSHMAN & WAKEFIELD

INDIAN FOOTPRINT

● Mumbai remained the most prominent data centre market in India in H12023, accounting for more than half of the total

- MUMBAI
- CHENNAI
- BENGALURU
- DELHI-NCR
- HYDERABAD
- PUNE
- KOLKATA

NOTE DATA IN %
SOURCE FROM
BYTES TO BUSINESS:
INDIA DATA CENTRE
MARKET POWERING
PROGRESS IN 2023;
CBRE SOUTH ASIA



The market leader for cloud and data centres globally, Amazon Web Services (AWS), is investing in renewable energy capacity for its data centres in India. “Since 2022, all the electricity consumed by our data centres can be attributed to renewable energy, thanks to Amazon’s investment in over 50 renewable energy projects to date across India, with an estimated capacity of more than 1.1 gigawatts,” says Ken Haig, Head of Energy and Environment Policy for AWS, Asia Pacific & Japan. Once all 50 wind and solar projects are operational, they are expected to generate 3.58 million megawatt hours (MWh) of power. That’s the equivalent of powering more than 1.1 million homes in New Delhi each year—according to the think tank Centre for Policy Research, an electrified house in Delhi consumes about 250–270 kWh of electricity a month on average.

KEEPING IT COOL

One other way of reducing the substantial electricity needs of data centres is through the cooling system. Earlier, air cooling and hot and cold aisle layout—where alternating rows of racks are placed between aisles that receive cold air from ACs and ones that have hot air exhausts—were popular. But there’s been a shift towards more sustainable cooling solutions, such as water-based chillers. This has been recognised as a significant contribution towards reducing the power required by cooling systems.

“Currently, the power usage effectiveness (PUE) of Indian data centres is estimated to be between 1.5 and 1.6. By adopting cutting-edge cooling technologies like direct liquid cooling and immersion cooling, it’s possible to lower the PUE to between 1.1 and 1.2,” adds Gupta of Yotta. The ideal PUE is 1, and the reading is higher if efficiency is lower. He adds that achieving such reductions in PUE would not only signify a decrease in power consumption but also mark a significant step towards greener, more environmentally friendly data centre operations.

There are other solutions, too. CtrlS Datacenters Ltd, which began operations in 2007 and now operates 15 data centres across eight key markets, has undertaken initiatives such as direct-to-chip cooling (DTC) and liquid cooling that are expected to reduce energy consumption by up to 40%, says Dillip Guru, Senior Vice President, Renewable Energy, CtrlS Datacenters. In DTC, non-flammable liquid is brought into contact with the chip through tubes. The liquid absorbs the heat and evaporates into vapour, thus cooling the equipment.

Natural ventilation and daylight, too, are being